Mathematical destination

Let μ replace A by AB and B by BA, simultaneously once per old tile. The rows $\mu^n(\mathrm{A})$ are nested prefixes; they define one endless word T, the Thue–Morse word. A *whole row* is exactly one full generation grown from a single A. A *crop* may start or end inside a replacement pair.

- **Pair invariant.** Number T from 0. True child-pairs start at even positions and have unequal letters. Equal neighbors therefore mark a gap between child-pairs.
- **Local theorems.** No genuine row contains `AAA`, `BBB`, `ABABA` or `BABAB`. Every genuine crop of at least five tiles has exactly one pairing into `AB`/`BA`, allowing at most one lone tile at each end. A shorter crop can have two immediate pairings.
- **Infinite theorem.** There are no $p > 0$ and $N \geq 0$ with $T[n + p] = T[n]$ for every $n \geq N$. The word is not eventually periodic. Equal neighbors rule out odd periods; decoding halves even periods. Page 4 supplies the proof.

Limits. Passing short tests does not certify a crop or a whole row. The repeating `ABBA` strip passes the four bans above. Finite repeats do occur in T; the claim concerns an endless repeating tail. Recovering pair alignment does not locate a crop's original position. Computation checks finite examples, not the infinite theorem.

Intended discoveries. Children grow and reverse the rule, distinguish whole rows from crops, and use equal neighbors to recover hidden seams. A successful first visit reaches Problems 1–4 with children choosing and checking. Problems 5–6 deepen the explanation; Problem 7 needs adult proof support unless children supply the arbitrary-period argument themselves.

Who uses this prototype

Start with the three children at the grades 4–5 table and the organizer: one pair, one referee, rotate after each row or claim. Prerequisites: read A/B strings, preserve order, and work with a partner; odd/even reasoning is needed only late. The four third graders may use the same packet in two pairs with their mathematician if they retain these choices. Their useful stopping point is Problem 4. Do not have an adult perform the decoding for children.

K–1, adult-led on the same pages. There is no K–1 packet, but the K–1 table can join the launch and then work with its parent, its own tile kit and page 1, with the adult reading. Grow A to 8 tiles (A, AB, ABBA, ABBABAAB), the child placing each pair under its old tile. Ask Problem 2 aloud: "Can three of the same letter ever sit together?" Play the fallback game below with 4-tile rows. Then build Problem 3's first and third rows (page 2) and undo them with tiles. Key in plain words: every new pair has one A and one B, so three of the same letter never sit together. The first row shrinks back to one A in three undos; the third has a pair BB, which no old tile makes, so it was not grown. It needs letters only, and no counting past 8. That K–1 children manage this is a prediction, untested.

Prepare and print

Per pair or rotating trio: one kit of 36 tiles, 18 A and 18 B, cut from week-76-materials.pdf (one card-stock page makes two kits; 20 mm tiles, about 10 minutes with a paper cutter), four small divider markers, two pencils, one handwritten key `A -> AB; B -> BA`, two blank recording sheets, and two loose row labels, "old" and "new." Two-colour counters at least 15 mm wide also work, lettered A on one face and B on the other: label both faces; color alone is insufficient. Allow 16 tile-widths plus working room per row (about 40 cm for 20 mm tiles). Separate child-pairs with small gaps; use markers locally. The largest build keeps 8 old and 16 new tiles at once. The printed boxes are references, not exact-fit tile mats.

Totals: for the target table, one kit and two spare dividers: 36 tiles, 6 dividers, 2 pencils, 1 key, 2 labels, 2 blank sheets. With the third-grade table added, prepare three kits and two shared spare dividers: 108 tiles, 14 dividers, 6 pencils, 3 keys, 6 labels, 6 blank sheets. A K–1 table adds one more kit, shared by its children (144 tiles, two printed tile pages, in all). Use a kit for the launch; no extra demonstration set is needed.

Print **students.pdf pages 1–4**, US Letter, single-sided at 100%: four sets for grades 4–5 (one per child plus one spare, 16 sheets). Add five sets for third graders (20 more sheets) if participating, and one copy of pages 1–2 for the parent at a K–1 table. Print one five-page guide per participating adult. Keep pages 2–4 back and issue one page at a time. No cutting or taping is required; isolate crops with fingers or spare paper. If cutting spare paper strips, supply one

pair of scissors per active table. Before use, rehearse one growth, one undo and one clipped crop with the actual tiles. **Physical rehearsal and classroom piloting remain unperformed.**

The hour

0–5 minutes: running around. **5–8:** short launch with all three tables. Put `BA` in the old row, leaving one tile-width between old tiles so each pair can sit under its parent, and let children handle the tiles. Say: "Each old tile makes one pair below it. A makes AB; B makes BA. Keep the old row until the checker agrees." Replace the old B with a separate BA pair. Ask one child to put AB below the old A. Point to the two parents and read the new `BAAB`. Say: "Stop. We have used each old tile once."

8–18: Problems 1–2. **18–28:** Problem 3. **28–40:** Problem 4 and, if ready, 5. **40–46:** Problem 6, a return to unfinished work, or the fallback below. **46–60:** sharing and clearing up, with slack for the moves between pages. Problem 7 can be a later visit; do not squeeze its proof into the final minute. Pause for movement if needed. These are planning estimates, not tested timings.

Fallback for either participating table: a child makes a 4- or 8-tile genuine row, then either leaves it alone or changes exactly one tile to the other letter. The partner checks whether it is the claimed whole row and gives a reason; swap roles. With three children, the referee records the maker's choice privately and checks the verdict. If growing itself is tiring, take turns making one legal replacement from the existing row. End with: "Which strip looked possible at first, and what finally settled it?"

Problem 1 Core, student page 1

Children grow from A to 16 tiles, keeping old and new rows separate, then collect every neighboring triple. The full growth history is

$$\mathrm{A} \longrightarrow \mathrm{AB} \longrightarrow \mathrm{ABBA} \longrightarrow \mathrm{ABBABAAB} \longrightarrow \mathrm{ABBABAABBAABABBA}.$$

The complete answer is `AAB, ABA, ABB, BAA, BAB, BBA`. The eight blank recording slots do not mean eight answers. A complete check slides a three-tile window one place at a time through the 14 possible starting places; repeated triples are recorded only once.

Hints in order. 1. "Which old tile are you replacing? Can your partner point to it?" 2. Slide two fingers around three neighboring tiles, moving one place each time. 3. Sort the recorded triples by first letter and check duplicates.

If struggling: stop at 8 tiles, preserve that row, and finish the 16-tile build together without taking over choices. All six triples already appear at length 8. Do not demand repeated copying of the growth history; one checked final row and a collection of triples suffice.

Problem 2 Core, student page 1

Neither `AAA` nor `BBB` can appear at any generation. Every stretch of three tiles contains a complete true replacement pair: either its first two tiles or its last two. That pair is AB or BA, so those two letters differ. A child can point to the complete pair in each of the two positions of a three-tile window.

Hints in order. 1. "Mark the pairs you just made. Where could three equal tiles sit?" 2. Move a three-tile window so it starts first on a pair, then halfway through one. 3. "In either place, is one whole pair trapped inside your window?"

If struggling: keep the pair markers and ask only about one attempted `AAA`. Skip the verbal generalization for now; record whether the child checked examples or explained both positions.

Problem 3 Core, student page 2

These are *whole-row* claims, so pair from the left edge with no lone tiles. Only the first claim is right. In printed order, the settling records are:

1. `ABBABAAB` → `ABBA` → `AB` → `A`: genuine.
2. `ABBAABBA` → `ABAB` → `AA`: the pair AA cannot decode.
3. `ABBABBAB`: its first split is `AB | BA | BB | AB`; BB fails immediately.
4. `ABBABAABABBABAAB` → `ABBAABBA` → `ABAB` → `AA`: fails at AA.

Hints in order. 1. "Show which two tiles came from the first old tile." 2. Decode one pair and compare with the replacement key. 3. "Does one successful undo reach A, or must you check an older row too?"

Matching also settles them. Each length has only one whole row, and the rows are nested, so comparing with the Problem 1 row settles every claim: claim 1 is its first 8 tiles, claims 2 and 3 differ from those at tiles 5 and 6, and claim 4 differs from the 16-tile row at tile 9. Accept it, then ask why a rejected row cannot shrink to A.

If struggling: start with claims 1 and 3, then 2 and 4. One bad pair settles a rejection. The worked example decodes to BAB; it makes no whole-row claim.

Hidden seams and forged strips

Problem 4 Core, student page 3

Children try both possible pair alignments. Parentheses below mark lone end tiles; bars separate complete pairs and end pieces. Decode *only complete pairs*. The full list, in page order, is:

Crop	All legal pairings	Old tiles kept
ABA	AB | (A) or (A) | BA	A or B
BAABA	BA | AB | (A)	BA
AABB	(A) | AB | (B)	A
ABBAABBA	AB | BA | AB | BA	ABAB

Why complete: either the first tile belongs to a pair, or it is lone. Each choice forces every later pair and the possible final lone tile. For BAABA, the other choice contains AA; for AABB, pairing at the first tile starts with AA; for ABBAABBA, leaving its first A alone starts with BB. Both choices work for ABA.

Important contrast: ABBAABBA really is a genuine crop, even though Problem 3 rejects it as a whole row. Its kept parent ABAB is also a crop; it need not decode from its own left edge all the way to a single A. No contradiction arises from a cut edge or a lone tile.

Hints in order. 1. "Could the cut have split the first pair? Try leaving the first tile alone." 2. "Can equal neighbors belong to one replacement pair?" 3. "Once one pair is placed, can the next pair move independently?"

If struggling: compare ABA with AABB, then stop. Use movable tiles and markers; a child may dictate the retained parent instead of copying it. The worked crop AABBA has (A) | AB | BA, keeping AB.

Problem 5 Further, student page 3

No genuine five-tile crop permits two pairings. An equal neighboring pair must straddle a seam, so it fixes the alignment everywhere in that crop. A five-tile crop without equal neighbors would have to be ABABA or BABAB; both are impossible.

Why those two are impossible: the first, third and fifth tiles have the same letter. They occupy the same half of three consecutive true child-pairs. If they are first halves, their parent letters are those letters; if second halves, their parent letters are the opposite letters. Either way the parent has AAA or BBB, forbidden by Problem 2. A child can extend the clipped end-pair with a tile to see the three equal parents. Adult help may be needed to account for the clipped pair.

Hints in order. 1. "Try the two alignments on your chosen five tiles. What breaks one?" 2. "What would a five-tile strip with no equal neighbors look like?" 3. For that alternating attempt, complete its clipped end-pair and inspect its three parents.

Stopping point: an explained seam in one crop is worthwhile. If children only test several crops, record a conjecture rather than an all-crops explanation. The proof above is the additional step.

Problem 6 Further, student page 4

Repeating AB: take the finite piece ABABA. It is forbidden by the preceding argument (BABAB also works).

Repeating ABBA: take its first ten tiles, ABBAABBAAB. Its BB forces the split

$$\mathrm{AB} \mid \mathrm{BA} \mid \mathrm{AB} \mid \mathrm{BA} \mid \mathrm{AB} \longrightarrow \mathrm{ABABA}.$$

If this ten-tile crop came from a genuine row, its five decoded letters would be a genuine parent crop. They are forbidden. This is a finite certificate, with no discarded or invented endpoint. The eight-tile ABBAABBA alone is *not* a certificate; it occurs genuinely.

Where the piece starts matters. From the strip's first or third tile, ten tiles reach a forbidden parent (BAABBAABBA $\to$ BABAB from the third). From the second or fourth, decoding only complete pairs needs eleven: one lone tile and five pairs (BBAABBAABBA $\to$ BABAB, AABBAABBAAB $\to$ ABABA). A child who reads the lone end tiles' parents, as in Problem 5 (a lone first tile came from the opposite letter, a lone last tile from the same letter), has a certificate at eight: BBAABBAA gives ABABA and AABBAABB gives BABAB. Accept either.

Hints in order. 1. "Mark a short piece of each repeating strip that you want to test." 2. "Which seams are forced? What do the complete pairs become?" 3. If eight tiles give only ABAB or BAB, extend the piece by one complete pair, or read the lone end tiles' parents, and decode again.

If struggling: settle the AB strip and postpone the other. Avoid arguing that a strip is false merely because it fails to shrink to A: that test applies to the whole-row claims in Problem 3.

The endless word and the adult proof

Problem 7 Adult-supported continuation, student page 4

Answer: no. First check that children have seen each row begin with the row before (A, AB, ABBA, ABBABAAB on page 1); "the endless strip" depends on it. Let children propose block lengths and show how marked seams behave under their shifts. Such experiments motivate the proof below; they cannot settle every length.

Hints in order. 1. "If this tail really repeats, what happens to an equal neighboring pair when you move it one full block?" 2. "What happens to its seam after an odd shift?" 3. "If the shift is even, keep the first tile in each true pair. How big is the shift now? Can you keep halving forever?"

Readiness and stopping: two marked copies of a short row can demonstrate a shift. If arbitrary lengths or the difference between a patch and an infinite tail is unclear, save this for another visit. Distinguish an adult-supplied proof from a child's explanation.

Mathematics behind the week

Definition and self-similarity. Since $\mu(\mathrm{A})$ begins with A, applying μ preserves the prefix relation; every generated row begins with the preceding one. The limiting word satisfies $T[2n] = T[n]$ and $T[2n+1] = \overline{T[n]}$ for $n \geq 0$, where the bar exchanges A and B. Thus reading the first letter of every true child-pair recovers T itself. This follows from the defining replacement, not a long inspection.

Local facts. A three-tile window always contains an unequal child-pair. An alternating five-tile window forces three equal consecutive parents in either alignment (Problem 5). This proves the four local bans. An equal pair then fixes the seams in every genuine five-tile crop.

No eventual period. Suppose $T[n+p] = T[n]$ whenever $n \geq N$, for positive p.

1. Equal neighbors occur arbitrarily far out: each aligned four-tile block is $\mu^2(\mathrm{A}) = \mathtt{ABBA}$ or $\mu^2(\mathrm{B}) = \mathtt{BAAB}$, whose middle letters agree. Such an equal pair starts at an odd index.
2. If p is odd, choose such a pair entirely beyond N. Shifting it by p preserves equality and puts its first letter at an even index, inside a true child-pair. But true child-pairs are unequal. Contradiction.
3. If $p = 2q$, then for every sufficiently large n, $T[n+q] = T[2n+2q] = T[2n] = T[n]$. So q is an eventual period of the same word; a changed finite starting point is harmless. Repeatedly halve an even positive period. A positive integer eventually becomes odd, contradicting step 2.

This proves Problem 7 for every proposed period and every finite discarded beginning.

Where next. Compare substitutions with ambiguous inverses, or investigate recurring finite patches. Order enforced across scales also matters in aperiodic tilings; this one-dimensional proof establishes no geometric tiling theorem.

Source lineage

Classical substitution facts; independently arranged packet instances and seam/period-halving presentation. Researcher-authored references:

- Richard G. Swan, *The Morse Sequence*, Lemma 2.2: https://www.math.uchicago.edu/~swan/expo/Morse.pdf
- Carl D. Offner, *Repetitions of Words and the Thue-Morse sequence*, §3: https://www.cs.umb.edu/~offner/files/thue.pdf
- Jean-Paul Allouche and Jeffrey Shallit, *The ubiquitous Prouhet-Thue-Morse sequence*, §§2–4: https://cs.uwaterloo.ca/~shallit/Papers/ubiq15.pdf

After the session

Record each table's stopping point, what children tried or said, a drawing or claim worth keeping, and any adult support. Label each claim *tried*, *guessed*, *checked in some cases*, or *explained*. Note handling or reading bottlenecks and where to resume. Classroom use remains unpiloted.